11.14 Parabola, Not Catenary

WORKSHEET

The question

STEP I 1991, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/91-s1-q10

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself At each joint A_i, three forces act: the tensions in the two neighbouring segments, and the weight w hanging from the strut. Which component of tension can the weight not affect?

STAGE 1 The equations at each joint

Covers: the equilibrium equations

Ask yourself At each joint, the weight is vertical. So what can you say about the horizontal components of the tensions in the two neighbouring segments? Resolve horizontally and vertically.

Pause and try At the middle joint, use the symmetry of the bridge. How much of its strut's weight does each of the two segments there support?

STAGE 2 Slopes from weights

Covers: the ratios

Ask yourself Work outwards from the middle. What is the vertical component of tension in each segment? Divide by the common horizontal component to get the slopes, and write each slope in terms of the heights y_i and h.

STAGE 3 **The parabola**

Covers: the parabola

Ask yourself Count joints outwards from the middle. Add up the drops from the middle joint: what is $1+3+5+\cdots+(2j-1)$?

What's really going on

Parabola or catenary?

The continuous version of the argument explains both curves. For a chain $y(x)$ under constant horizontal tension H, the vertical component of tension is Hy', and over a short stretch it changes by the load carried there: $H\,\mathrm{d}y' = \mathrm{d}W$.

- If the load is spread evenly *horizontally* (a heavy roadway, weight ρ per unit length of road), then $Hy'' = \rho$, so y is a quadratic: a parabola.
- If the load is the chain's own weight, spread evenly *along the chain* (ρ per unit length of chain), then $Hy'' = \rho\sqrt{1+y'^2}$, whose solution is the catenary $y = c\cosh\frac{x}{c}$, with $c = H/\rho$.

Near the bottom, where the chain is nearly horizontal, the two are almost the same; they differ near the supports, where the chain is steep and a given horizontal distance contains more chain. Real suspension bridges, whose deck is much heavier than their cables, are very close to parabolas.

Upside-down chains and arches

A chain is in pure tension; turn its shape upside down and you have an arch in pure compression, with no tendency to bend. Robert Hooke published this in 1675 as an anagram, which decoded reads: "As hangs the flexible line, so but inverted will stand the rigid arch." Christopher Wren and Hooke used the idea for the dome of St Paul's Cathedral, and Antoni Gaudí designed the Sagrada Família in Barcelona by hanging weighted strings from the ceiling and photographing them upside down. The Gateway Arch in St Louis is a (weighted) inverted catenary.

The catenary challenge

In 1690 Jakob Bernoulli challenged mathematicians to find the shape of a hanging chain. Within a year, solutions were published by his brother Johann, by Leibniz and by Huygens. Johann Bernoulli later recalled with glee that his brother, who had set the problem, had failed to solve it. The catenary was one of the first triumphs of the new calculus, and it led to the hyperbolic functions cosh and sinh, studied systematically by Vincenzo Riccati and Johann Heinrich Lambert in the eighteenth century.

Try this next

1. **An even number of struts.** Repeat the question with $2n$ struts, so that the lowest point of the chain is a horizontal segment in the middle. Do the joints still lie on a parabola?

2. **The tension.** Find the greatest tension in the chain, in terms of H, w and n. Where is it?

3. **The catenary.** Verify that $y = c\cosh\frac{x}{c}$ satisfies $y'' = \frac{1}{c}\sqrt{1+y'^2}$. Show that the length of the catenary from $x = 0$ to $x = X$ is $c\sinh\frac{X}{c}$.

4. **Galileo's error.** A chain of length $2L$ hangs between two supports at distance $2d$ apart. For L only slightly larger than d, show that the sag of the catenary is approximately the same as that of a parabola. Why did Galileo's experiments not reveal the difference?

Further reading

- **E. H. Lockwood, *A Book of Curves*** (Cambridge University Press). The catenary, the parabola and their relatives.
- **Jacques Heyman, *The Stone Skeleton*** (Cambridge University Press). How masonry arches and domes stand up, from Hooke's chain onwards.
- **David Billington, *The Tower and the Bridge*** (Princeton University Press). The art and engineering of bridges.

The full guided solution is in Section 11.14:
https://beyondstep.keithmcnulty.org/ch11/bridge-parabola.html